

ReTrace: the DLM is both generator and keyed reconstruction probe

one block at a time

1 Reference DLM

keep the native block schedule

Current state



diffusion LM

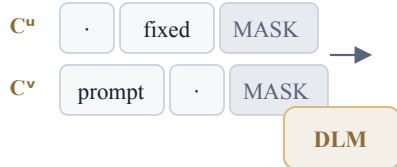
candidate	p_θ
the	.42
a	.25
this	.13

sample once and commit

2 Response Bank + Gate

RRB records; BCG selects

Mask block + suffix; ablate the fixed prefix



RRB entry $g_i(z) = \log p^v - \log p^u$

token	g_i	sign
the	+ .8	+
a	- .6	-
this	+ .1	+

$q_+ = .53$

$q_- = .47 \Rightarrow \text{carrier}$



3 RAR embedding

response-aligned resampling

keyed orientation

$$w_i \epsilon_i = +1$$

Draw from the unchanged live row

try	token	p_i	g	decision
1	a	.25	-	reject
2	the	.42	+	accept

accept if sign matches and $p_i \geq \kappa \max p_i$

after R misses: one unchecked fallback draw

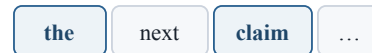
Committed block



4 RRV verification

replay, count, test

Finished text



re-mask and rebuild
the same Response Bank

Carrier matches m_i



$$T = \sum_i m_i$$

$$p_{\text{det}} = \Pr[\text{Bin}(n, 1/2) \geq T]$$

detected

REPRODUCIBLE table uses only the fixed prefix

MODEL-PROPOSED retries reuse one probability row

EXACT NULL conditional Binomial($n, 1/2$)